Lecture 11: Gas & Ice Giants
Jupiter, Saturn, Uranus, Neptune

Planetary Systems

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Jupiter’s south pole, JunoCam. Credit: NASA/JPL-Caltech/SwRI/MSSS

Learning Objectives

By the end of this lecture, you will be able to:

- Compare the interiors and atmospheres of the four giant planets
- Derive the Roche limit and apply it to Saturn's rings
- Explain why the giant-planet moons host oceans, and use the ice giants as analogues of the commonest exoplanets

Recap: Where We Are in the Course

- **L9 (Earth & Venus):** rocky-planet near-twins with divergent surface evolutions
- **L10 (Mercury & Mars):** rocky-planet limiting cases (dense core vs. volatile-rich)
- **Today:** the four giant planets, >99.5% of all planetary mass beyond the Sun

Today: How do gas and ice giants work, and why do their moons host oceans?

1. Gas Giants and Ice Giants: An Overview

Solar system family portrait, Voyager 1. Credit: NASA/JPL-Caltech

Two Classes, Four Planets

- **Gas giants** (Jupiter, Saturn): H/He envelopes dominate the mass (318 and 95 $M_\oplus$), together ~ 92% of all planetary mass
- **Ice giants** (Uranus, Neptune): H/He only ~ 10–20% of the mass (14.5 and 17.1 $M_\oplus$); the bulk is water, ammonia and methane fluids
- No solid surface (the 1 bar level is the reference); rapid rotation (~ 10 h) makes them oblate by 6–10%

Energy Budgets: Internal Heat Excess

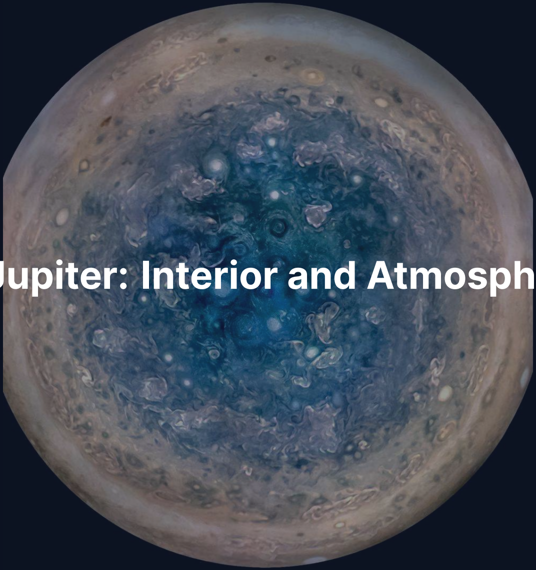
- **Jupiter (**1.7×**):** slow Kelvin-Helmholtz gravitational contraction
- **Saturn (**1.8×**):** gravitational energy release from helium rain
- **Neptune (**2.6×**) vs. Uranus (**1.06×**):** strong internal flux against near solar equilibrium

Three of the four giants radiate far more than they receive from the Sun; Uranus is the anomalously cold exception.

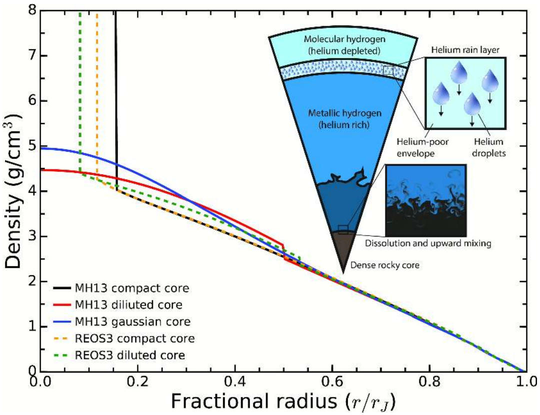
Why Rapid Rotation Matters

- **Rapid spin:** Jupiter (9 h 56 min) and Saturn (10 h 33 min) rotate in ~ 10 h
- **Zonal banding:** Coriolis forces organise the circulation into alternating jets
- **Gravity harmonics J_n:** oblateness and deep flows encode the mass distribution, measured by Juno and Cassini

2. Jupiter: Interior and Atmosphere



Jupiter’s south pole, JunoCam. Credit: NASA/JPL-Caltech/SwRI/MSSS



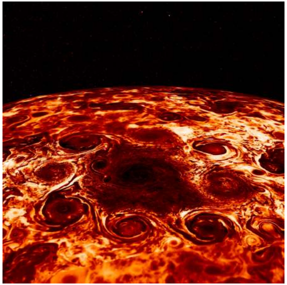
Density against fractional radius for Jupiter interior models with a compact core (dashed) and a dilute core; metallic hydrogen begins at ~ 100 GPa, $r/R_J \approx 0.85$. Wahl et al. (2017).

Juno's gravity harmonics J_4 to J_{10} rule out a compact ~ 10 $M_\oplus$ core: 20–40 $M_\oplus$ of heavy elements spread over the inner 30–50% of the radius, mixed during accretion or by a giant impact (Liu et al. 2019).

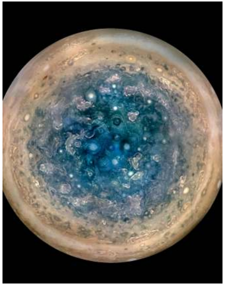
Crescent Jupiter and the Great Red Spot, JunoCam (perijove 3, December 2016). Credit: NASA/JPL-Caltech/SwRI/MSSS / R. Tkachenko.

Alternating zonal cloud belts bracket the anticyclonic Great Red Spot, which has shrunk from ~ 40,000 km in 1900 to ~ 14,000 km today.

Polar Cyclones



North pole: 1 + 8 cyclones, Adriani et al. (2018). Credit: NASA/JPL-Caltech/SwRI/ASI/INAF/JIRAM



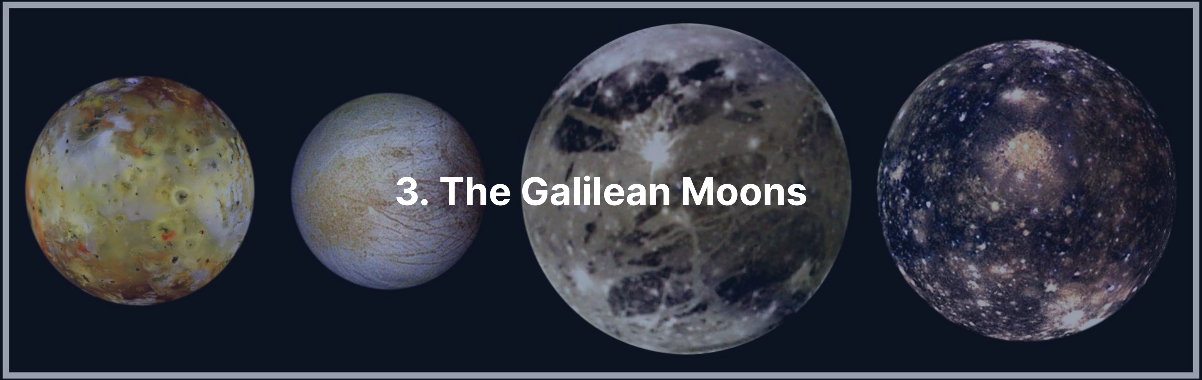
South pole: 1 + 5 cyclones. Credit: NASA/JPL-Caltech/SwRI/MSSS

Stable polygonal arrangements persist across many Juno orbits; polar weather was not seen at high resolution before 2016.

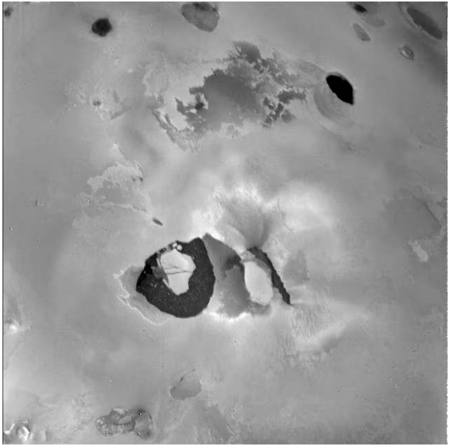
Jets, Aurorae, and the Great Blue Spot

- **Jet penetration:** winds extend to ~ 3000 km before magnetic braking enforces rigid rotation (Kaspi et al. 2018)
- **Io mass loading:** ~ 1 $\mathrm{t\,s^{-1}}$ of sulfur and oxygen ions feed the strongest aurorae in the solar system; UV footprints mark the field lines to Io, Europa and Ganymede
- **Great Blue Spot:** isolated equatorial magnetic anomaly mapped by Juno (Connerney et al. 2022)

3. The Galilean Moons

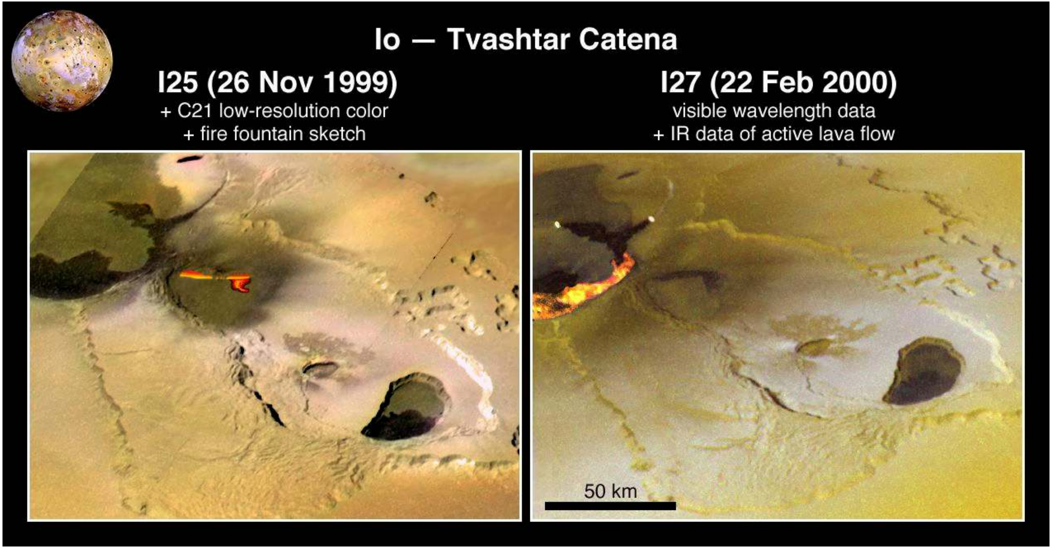


The Galilean moons at correct relative size: Io, Europa, Ganymede, Callisto. Credit: NASA/JPL/DLR

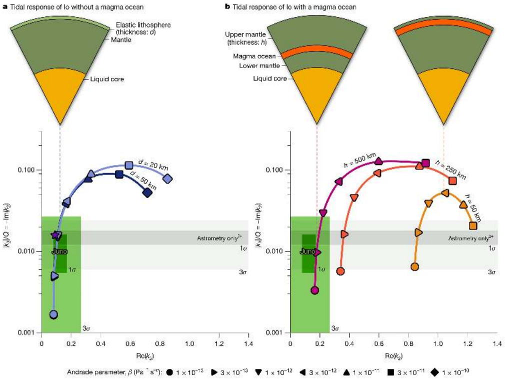


Loki Patera lava lake on Io, Voyager 1 (1979). Credit: NASA/JPL.

About 400 active volcanic centres and ~ 10^{14} W of tidal heat driven by the 1:2:4 Laplace resonance, predicted by Peale et al. (1979) days before Voyager 1; the surface is resurfaced on ~Myr timescales.

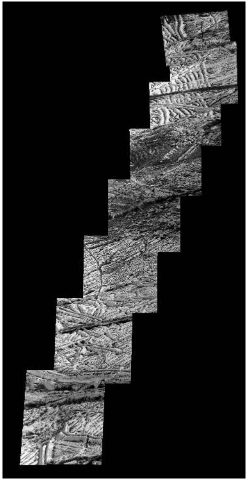


The Tvashtar Catena fire fountain on Io in a Galileo composite of orbits C21, I25 (26 November 1999) and I27 (22 February 2000). Credit: NASA/JPL/University of Arizona.



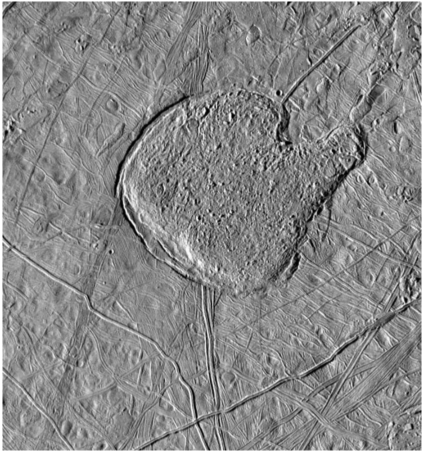
Io's measured tidal response, $|k_2|/Q$ against $\mathrm{Re}(k_2)$, for interior models without (a) and with (b) a global magma ocean; green boxes are the Juno constraints. Reproduced from Park et al. (2025).

Models without a magma ocean pass through the Juno box, while a magma ocean shallower than about 500 km gives a far larger $\mathrm{Re}(k_2)$ than measured: the data favour a mostly solid mantle.



Europa surface mosaic from the Galileo orbiter. Credit: NASA/JPL-Caltech/SETI Institute.

Global lineae fractures and disrupted chaos terrain reflect ongoing tidal heating; the surface is only 40–90 Myr old.

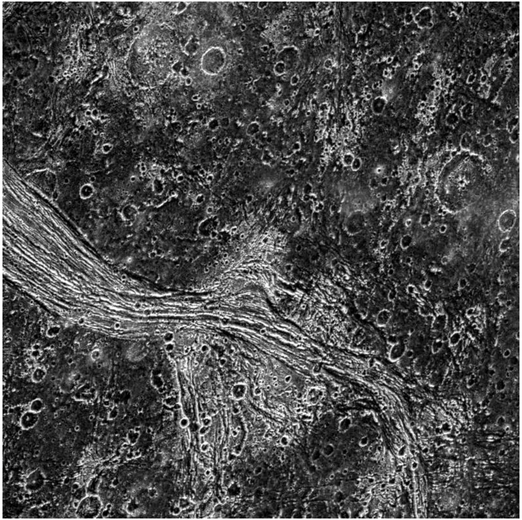


Conamara Chaos on Europa, Galileo (PIA01640). Credit: NASA/JPL-Caltech/ASU.

Chaos morphology records episodic basal melting, and the induced dipole seen by the Galileo magnetometer requires a global conducting ocean (Khurana et al. 1998): 6–25 km of ice over a ~ 100 km ocean, about one Earth-ocean volume.

Ganymede imaged by JunoCam during the 7 June 2021 close flyby. Credit: NASA/JPL-Caltech/SwRI/MSSS.

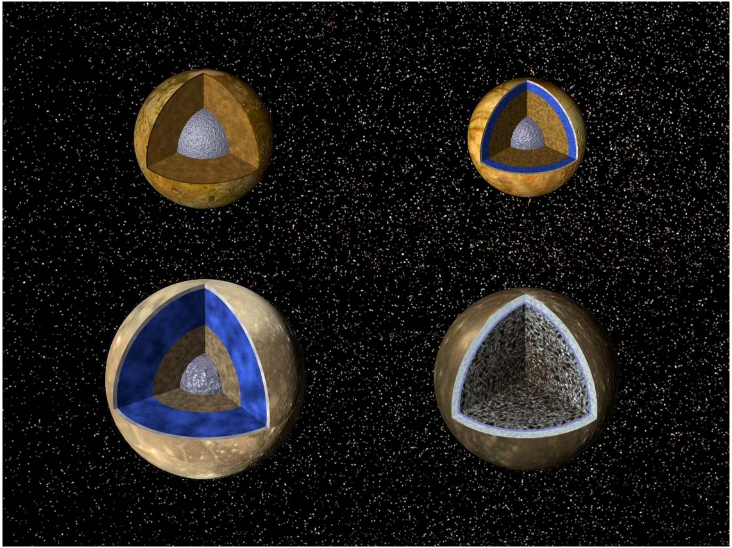
Radius 2634 km, larger than Mercury; the only moon with an intrinsic dynamo (Galileo 1996); fully differentiated, with a ~ 100 km ocean below a ~ 150 km ice shell (Saur et al. 2015).



Grooved bright terrain (Lagash Sulcus) cutting ancient cratered dark terrain in Marius Regio on Ganymede, Galileo, 6 June 1997, about 288 m per pixel. Credit: NASA/JPL-Caltech/Brown University.

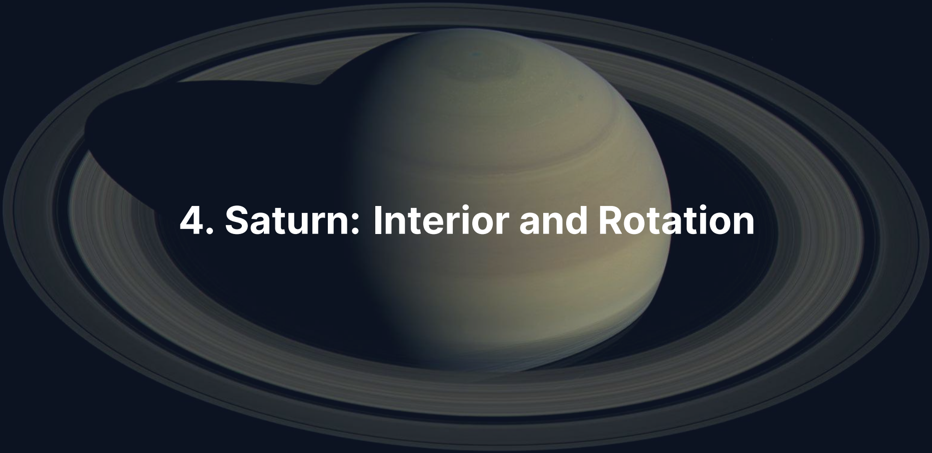
Callisto in global colour view, assembled from Galileo and Voyager images. Credit: NASA/JPL-Caltech.

R = 2410 km, density 1834 $\mathrm{kg\,m^{-3}}$, $C/MR^2 \approx 0.355$: only partially differentiated (Anderson et al. 2001), outside the Laplace resonance and without tectonic resurfacing, yet an induced field points to a subsurface ocean.



Interior structures of the four Galilean moons at comparable scale: only Callisto (right) keeps an undifferentiated rock-ice interior. Credit: NASA/JPL-Caltech.

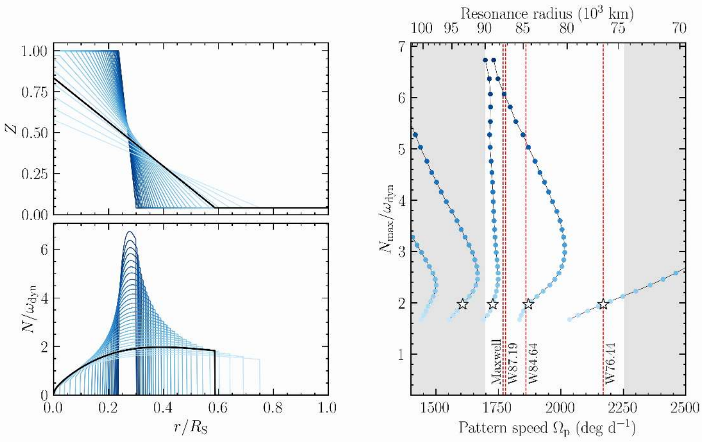
4. Saturn: Interior and Rotation



Saturn and its rings, Cassini (PIA21046). Credit: NASA/JPL-Caltech/SSI

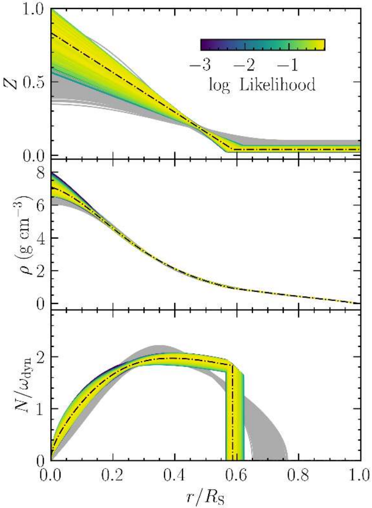
Saturn Interior: Lower Pressures, Helium Rain

- Mass 95 $M_\oplus$, central pressure ~ 1/3 of Jupiter's: the molecular-to-metallic hydrogen transition sits deeper, at milder conditions
- Helium becomes immiscible in metallic H at 1–3 Mbar, 5–10 kK; the falling droplets release gravitational energy and close Saturn's heat budget
- The rain depletes the upper envelope in helium, consistent with the Voyager observations; the Cassini Grand Finale (2017) mapped the gravity field

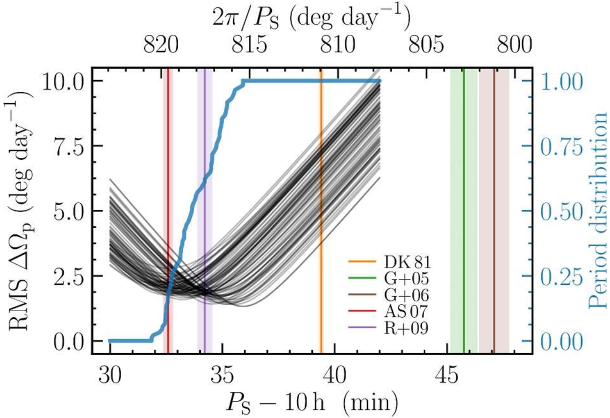


Kronoseismology: f-mode resonances of Saturn's interior imprinted as waves in the C ring.
Mankovich & Fuller (2021).

The ring waves require a stably stratified dilute core out to ~ 60% of R_S holding ~ 17 $M_\oplus$ of heavy elements.

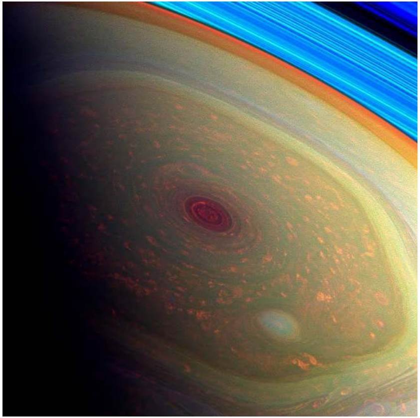


Saturn’s heavy-element fraction $Z(r)$ (top), density (middle) and Brunt-Vaisala frequency (bottom) against fractional radius, coloured by model likelihood, maximum-likelihood profile in yellow. Reproduced from Mankovich & Fuller (2021).



RMS pattern-speed residual between interior models and the observed C-ring density waves against the assumed rotation period. Mankovich et al. (2019).

Saturn's nearly axisymmetric dipole gives no radio period; the C-ring f-mode frequencies give P_S = 10 h 33 min 38 s and resolve the decades-long Voyager-Cassini ambiguity.



Saturn's hexagonal polar jet, Cassini false-colour view (PIA14946). Credit: NASA/JPL-Caltech/SSI/Hampton University.

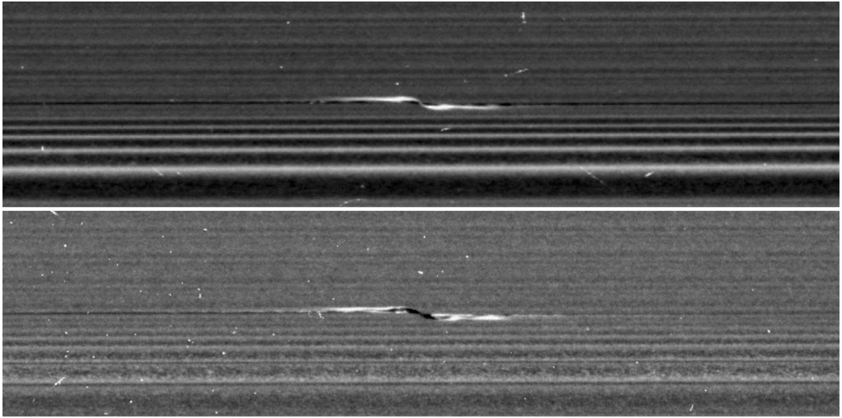
The same $\mathrm{NH_3/NH_4SH/H_2O}$ cloud decks as Jupiter, deeper and muted; the hexagon at ~ 78° N is a six-sided standing Rossby wave seen since Voyager (1980 to 1981), and the equatorial jet reaches ~ 400 $\mathrm{m\,s^{-1}}$.

5. Saturn’s Rings

Saturn’s main ring system, Cassini (PIA21046). Credit: NASA/JPL-Caltech/SSI

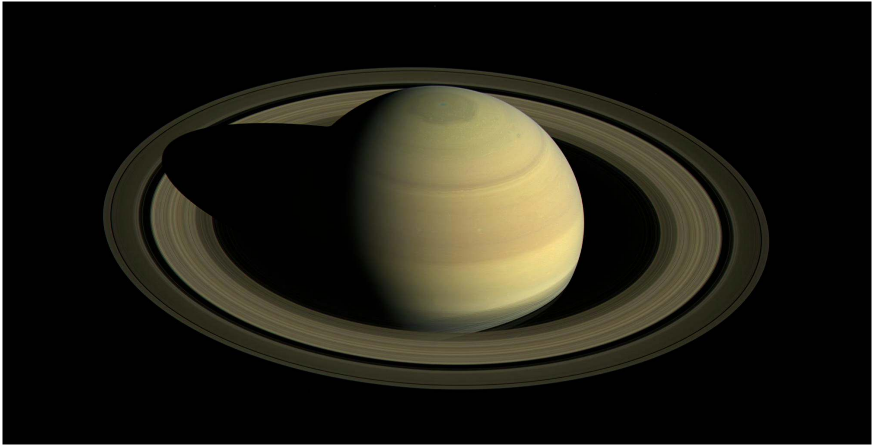
Natural-colour radial scan across the C ring, B ring, Cassini Division and A ring, Cassini (PIA08389). Credit: NASA/JPL-Caltech/SSI.

The rings span 67,000–140,000 km, consist of > 95% pure water ice, and are only ~ 10 m thick.



Two Cassini close-ups of the same propeller feature in the A ring, the wake of an embedded ~ 1 km moonlet. Credit: NASA/JPL-Caltech/Space Science Institute.

Prometheus and Pandora shepherd the F ring, Pan clears the Encke Gap, and the propeller moonlets show orbital migration directly within a disk (Tiscareno et al. 2013).



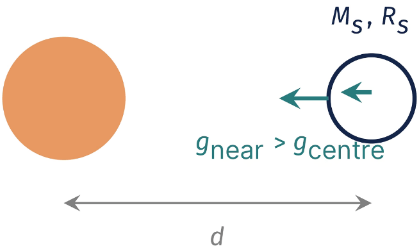
Saturn and its rings, Cassini. Credit: NASA/JPL-Caltech/SSI.

Only ~ 1.5×10^{19} kg, ~ 40% of Mimas (Iess et al. 2019), and still bright despite micrometeoroid infall: about 100 Myr old (Crida et al. 2019), draining as ring rain on 10^8 yr scales, formed by the tidal disruption of an icy moon (Wisdom et al. 2022).

6. Blackboard Derivation: The Roche Limit

Solar system family portrait. Credit: NASA/JPL-Caltech

Goal and Strategy



Goal: the distance inside which tidal stretch beats self-gravity (Roche 1849): rings persist, moons cannot accrete.

- Planet: mass M_p, radius R_p, mean density ρ_p
- Satellite: mass M_s, radius R_s, density ρ_s, at distance $d \gg R_s$
- Balance the tidal stretch $\Delta a_{\text{tidal}} \approx 2GM_pR_s/d^3$ against the surface self-gravity GM_s/R_s^2

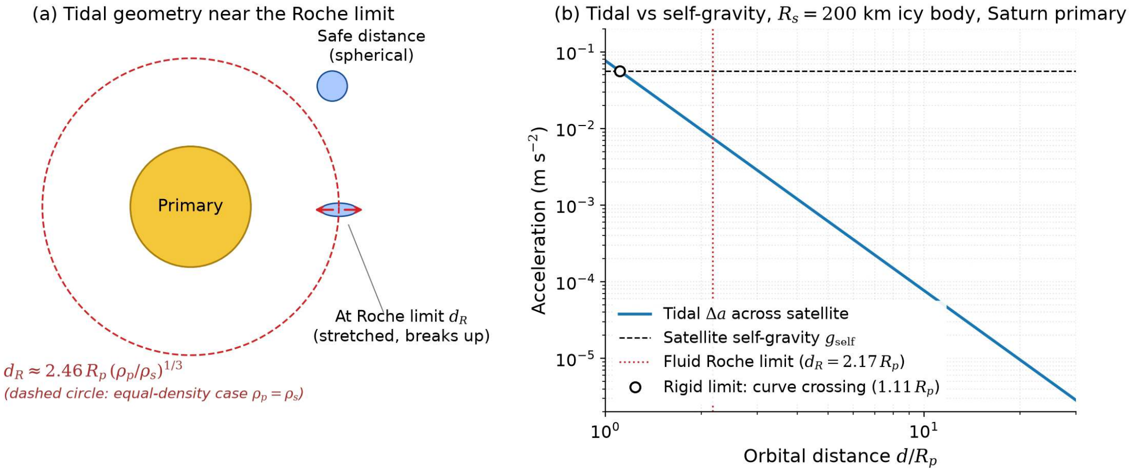
To the board.

Fluid vs. Rigid Roche Limit

- Rigid case: the body holds its spherical shape until disruption: $d_R^{\text{rigid}} = 2^{1/3} R_p(\rho_p/\rho_s)^{1/3} \approx 1.26\, R_p(\rho_p/\rho_s)^{1/3}$
- Fluid case: the body deforms into an ellipsoid first and disrupts further out; Roche's calculation gives the prefactor ≈ 2.46:

$$\boxed{d_R \approx 2.46\, R_p \left(\frac{\rho_p}{\rho_s}\right)^{1/3}}$$

Real bodies with material strength sit between the rigid and fluid limits.



Roche limit geometry: (a) a fluid satellite elongates before it disrupts at d_R; (b) differential tidal acceleration against surface self-gravity for an icy body near Saturn. Course figure.

Apply to Saturn's Rings

$$R_p = 58{,}232\ \mathrm{km}, \quad \rho_p = 687\ \mathrm{kg\,m^{-3}}, \quad \rho_s \approx 1000\ \mathrm{kg\,m^{-3}}$$

$$d_R \approx 2.46 \times 58{,}232 \times (0.687)^{1/3}\ \mathrm{km} \approx 2.46 \times 58{,}232 \times 0.883$$

$$\boxed{d_R \approx 126{,}000\ \mathrm{km}}$$

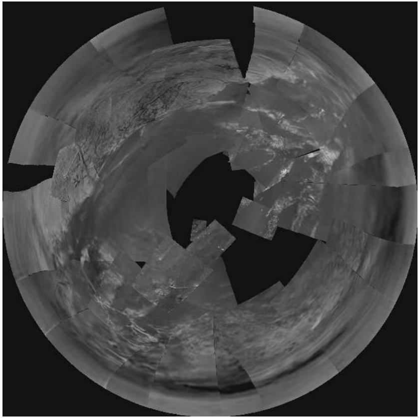
Within ~ 10% of A-ring outer edge (~ 137,000 km). Discrepancy reflects particle porosity, ice rigidity, and resonant structures.

Break

Earthrise, Apollo 8 (1968). Credit: NASA

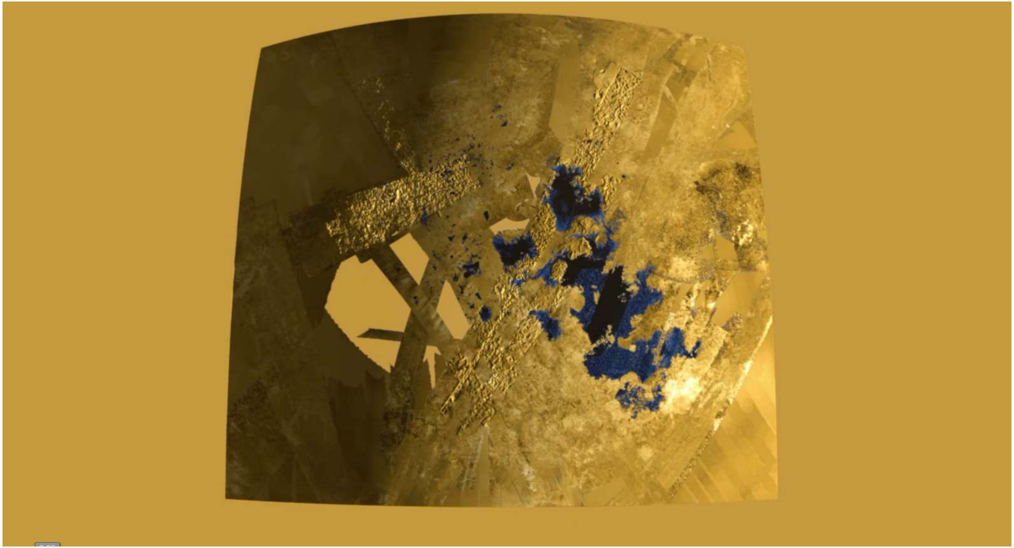
7. Saturn’s Moons: Titan, Enceladus, and More

Titan’s polar lakes, Cassini RADAR. NASA/JPL-Caltech/ASI/Cornell

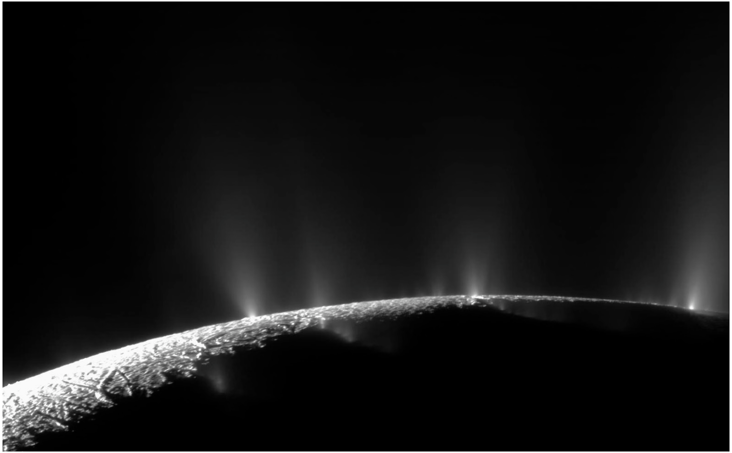


Huygens descent imagery of Titan's surface, 14 January 2005 (PIA07870). Credit: ESA/NASA/JPL-Caltech/University of Arizona.

R = 2575 km, the second largest moon; an $\mathrm{N_2}$ atmosphere at 1.5 bar (4× Earth's column density) and a 94 K surface near methane's triple point drive a methane cycle of clouds, rain, rivers and polar hydrocarbon lakes.



Cassini radar map of Titan’s polar hydrocarbon seas, including Ligeia Mare and Kraken Mare, dark and radar-smooth. Credit: NASA/JPL-Caltech/ASI/Cornell.

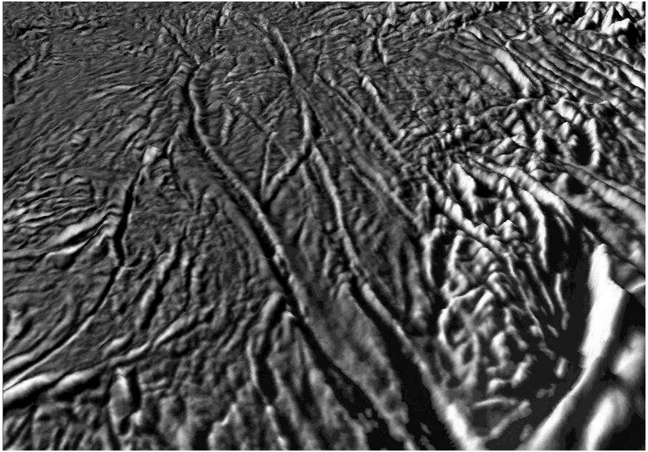


Cryovolcanic geyser jets erupting from the south-polar tiger stripe fractures of Enceladus, Cassini. Credit: NASA/JPL-Caltech/SSI.

Dozens of jets powered by > 10 GW of tidal heat spray subsurface ocean water directly into space: the most accessible ocean world.

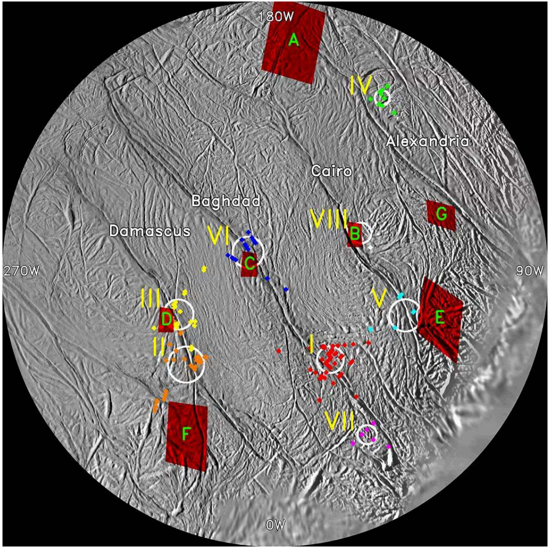
Enceladus: Habitability Ingredients

- Global subsurface ocean beneath a 20–30 km ice shell; the 2:1 resonance with Dione forces the eccentricity that powers the tidal heating
- Plume H_2 (Waite et al. 2017) and silica nanoparticles (Hsu et al. 2015): chemical disequilibrium from active rock-water reactions at hydrothermal vents
- Phosphates (Postberg et al. 2023): an essential biochemical building block



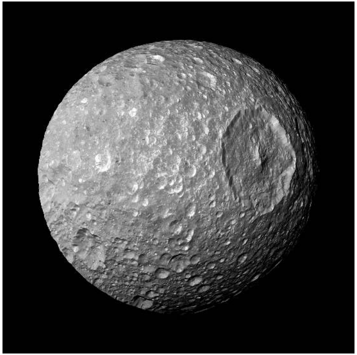
The four parallel tiger stripe fractures near the south pole of Enceladus that feed the plumes. Credit: NASA/JPL-Caltech/SSI.

The fractures are warmer than the surrounding terrain by tens of kelvins, and their ages and orientations track the stress field of the eccentric orbit around Saturn.

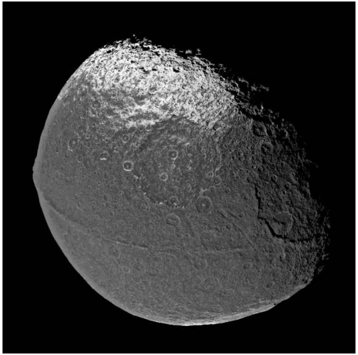


Cassini CIRS thermal map of the south polar terrain of Enceladus: the endogenic heat flow concentrates along the four tiger stripes. Credit: NASA/JPL-Caltech/GSFC/SwRI/SSI.

Mid-Sized Moons: Mimas and Iapetus

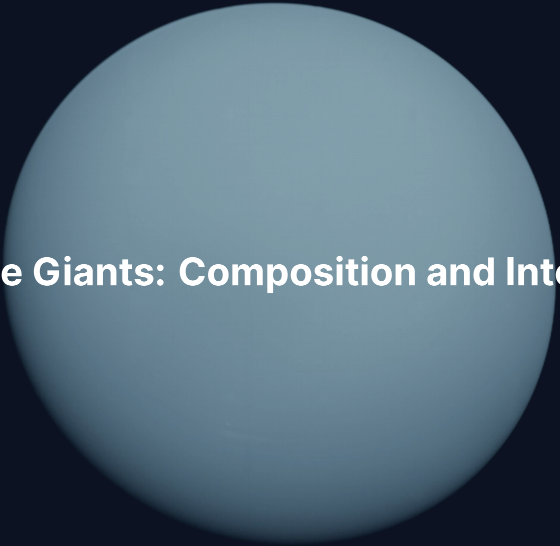


Mimas, Herschel crater (near the disruption threshold). Credit: NASA/JPL-Caltech/Space Science Institute



Iapetus, two-toned hemispheres. Credit: NASA/JPL-Caltech/Space Science Institute

8. Ice Giants: Composition and Interior

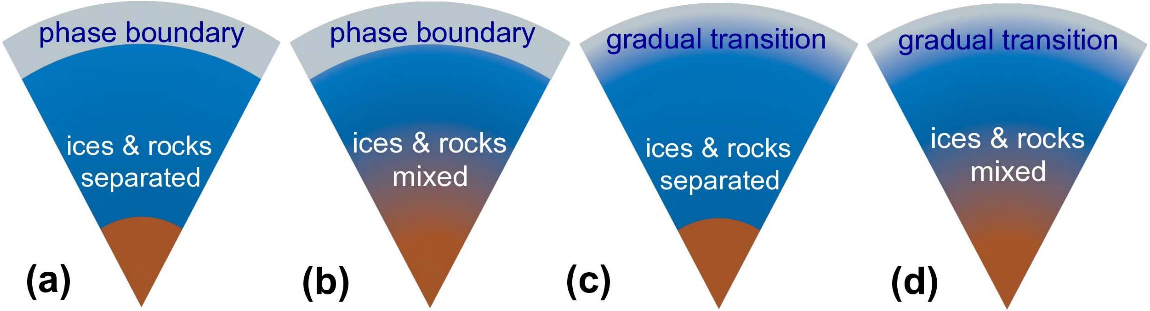


Uranus, Voyager 2 (1986). NASA/JPL-Caltech

Ice Giants: The Exotic Twins

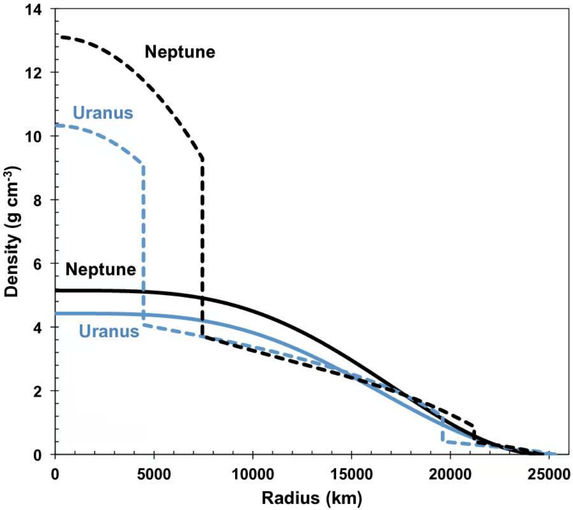
- Uranus: 14.5 $M_\oplus$, 4.0 $R_\oplus$; Neptune: 17.1 $M_\oplus$, 3.9 $R_\oplus$
- H/He envelope only 10–20% of the mass; the bulk is water, ammonia and methane in a compressed fluid state
- One spacecraft visit each: Voyager 2 in January 1986 and August 1989

The most under-explored major planets: the data are 35–40 yr old.



Possible ice-giant interiors, from sharp layer boundaries (a) to a fully gradual composition gradient (d). Helled et al. (2020).

The Voyager J_2 and J_4 cannot break the degeneracy between these density profiles.

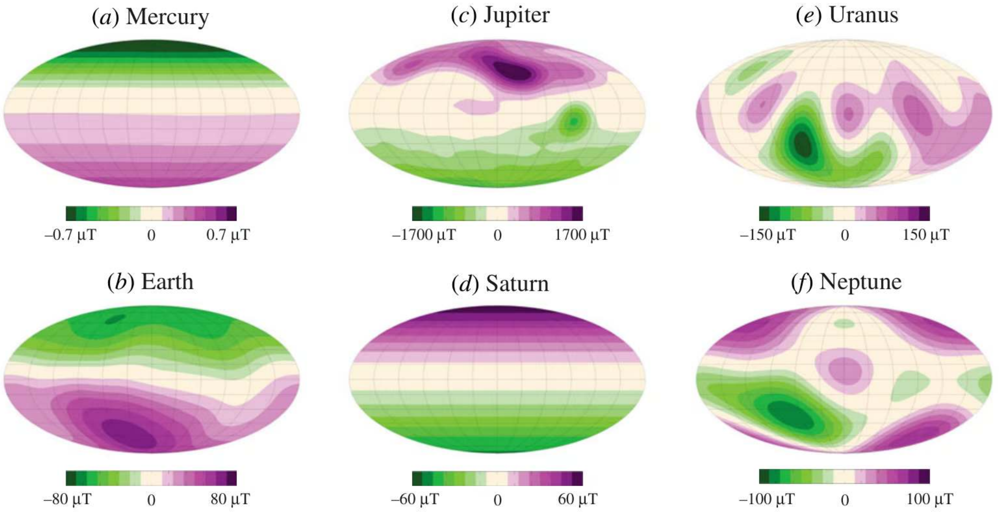


Density against radius for Uranus (blue) and Neptune (black): empirical profiles (solid) and three-layer models (dashed) match the gravity data equally well. Reproduced from Helled et al. (2020).

Superionic Ice and Tilted Dynamos

- **Superionic ice:** at 100–400 GPa a solid oxygen lattice with mobile protons conducts (Millot et al. 2019)
- **Tilted, offset fields:** 59° (Uranus) and 47° (Neptune), offset $\sim 1/3\,R_p$ and $\sim 1/2\,R_p$ from the centre
- **Thin-shell dynamos:** convection in an outer shell explains the multipolar geometry

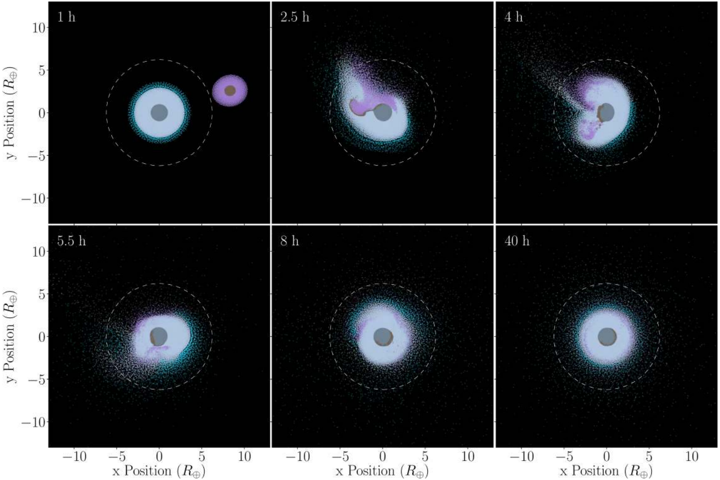
Superionic ice powers dynamos without metallic hydrogen.



Surface radial magnetic field of Mercury, Earth, Jupiter, Saturn, Uranus and Neptune, each on its own colour scale: aligned dipoles at Jupiter and Saturn, tilted and offset multipoles at Uranus and Neptune. Reproduced from Soderlund & Stanley (2020).

9. Uranus: The Tilted Planet

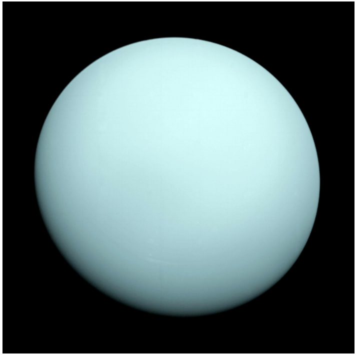
Uranus clouds, Voyager 2 (PIA18182)



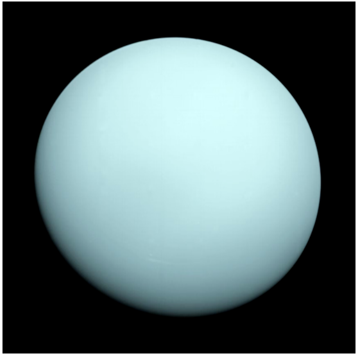
SPH simulation of a giant impact on proto-Uranus. Kegerreis et al. (2018).

An oblique collision tilted the spin axis to 97.8°, giving 42-yr polar seasons, and shed an equatorial debris disk.

Uranus: Featureless at Voyager, Active Now

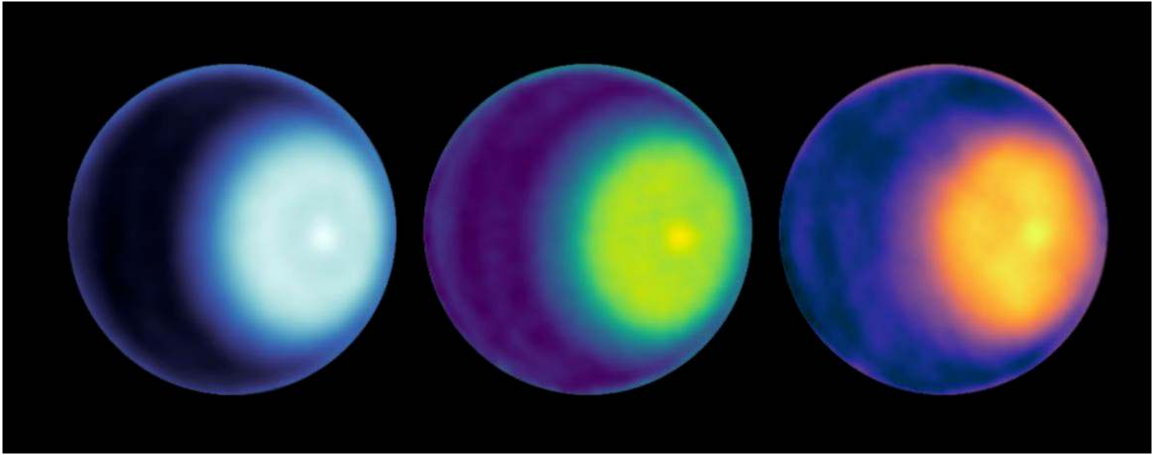


Uranus, Voyager 2 (1986). Credit: NASA/JPL-Caltech



Enhanced cloud bands, Voyager 2 (PIA18182). Credit: NASA/JPL-Caltech

Voyager 2 (1986) observed Uranus near solstice: pole-on orientation and muted cloud bands.



Uranus with its north-polar cyclone in recent observations.

Since the 2007 equinox, toward northern summer (2030), cloud activity and storms increase: a 2014 storm system from the ground, vivid rings and a polar cap from JWST (2023), and a north-polar cyclone confirmed by the VLA and JWST.

Uranus: Why So Cold Inside?

- **Heat anomaly:** ~ 1.06× the absorbed solar flux (Pearl et al. 1990); Neptune ~ 2.6×
- **Trapped heat:** a giant impact may have left a stably stratified interior that suppresses convection
- **Latent heat:** deep condensation of volatiles could delay the cooling

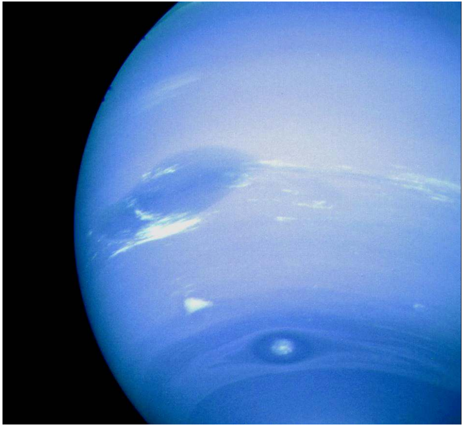
Uranus emits almost no heat beyond the sunlight it absorbs; Neptune does.

10. Neptune and Triton

Neptune Great Dark Spot, Voyager 2 (1989). NASA/JPL-Caltech

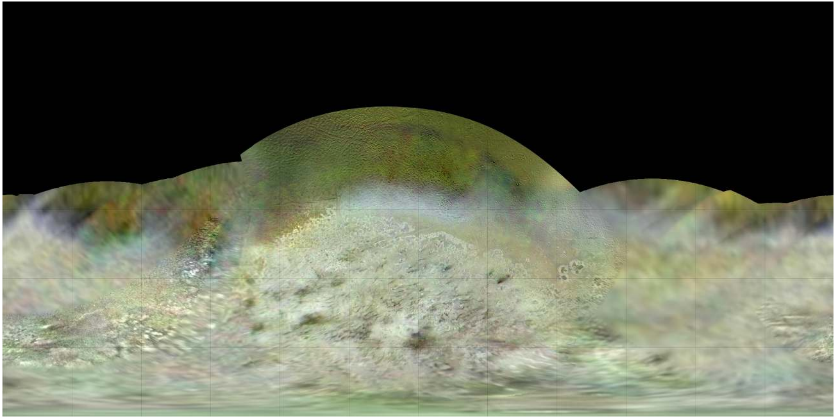
Neptune’s Great Dark Spot, Voyager 2 (1989). Credit: NASA/JPL-Caltech.

An anticyclonic storm the size of Earth, bordered by bright methane cirrus, seen during the 1989 flyby and gone by 1994 (HST); new dark spots emerge periodically.



The bright "Scooter" cloud feature near 42°S on Neptune, Voyager 2 (1989). Credit: NASA/JPL-Caltech.

The Scooter drifts westward relative to the 16.1 h interior rotation and the equatorial jet peaks at ~ 400 $\mathrm{m\,s^{-1}}$ on 1/900 of Earth's insolation: the winds draw on the ~ 2.6× internal heat excess (Pearl & Conrath 1991).



Triton's southern hemisphere, Voyager 2 mosaic, with cantaloupe terrain and dark geyser streaks. Credit: NASA/JPL-Caltech.

A captured Kuiper Belt object: regular moons form prograde, so the retrograde orbit ($i \approx 157^{\circ}$) means capture; active 8-km $\mathrm{N_2}$ geysers and a surface younger than $\lesssim 100$ Myr.

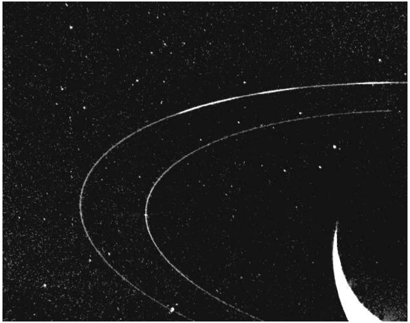
11. Ice Giant Rings

Neptune ring system, Voyager 2 (1989). NASA/JPL-Caltech

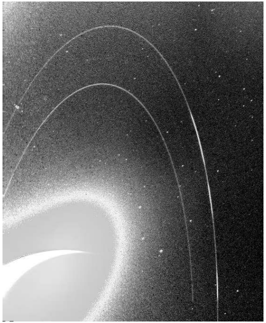
Ice Giant Rings: Narrow, Dark, and Carbonaceous

- **Occultation discovery:** Uranus's rings were found in 1977 by stellar occultation (13 narrow rings known today)
- **Compositional contrast:** Saturn's rings are > 95% bright water ice, ice-giant rings are dark organics
- **Disruption origin:** collisional destruction of small inner moonlets rather than a single large icy parent

Neptune Rings: Arcs Held by Resonance



Adams ring + Le Verrier ring. Credit: NASA/JPL-Caltech



Long-exposure forward-scattered view. Credit: NASA/JPL-Caltech

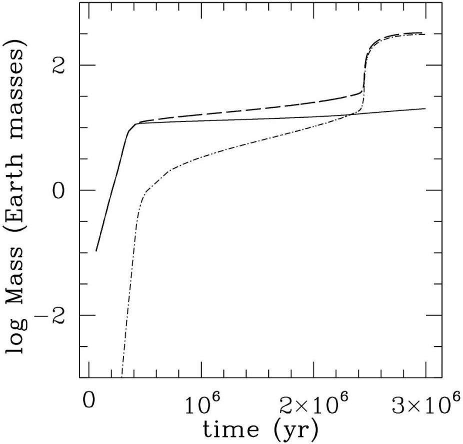
The Adams ring holds 5 distinct *arcs*, trapped by a mean-motion resonance with Galatea.

12. Comparative Payoff and Exploration Frontier

Europa Clipper concept. NASA/JPL-Caltech

Why Gas Giants and Ice Giants Diverged

- **Timing:** Jupiter and Saturn reached runaway gas accretion while the nebula was still massive; Uranus and Neptune grew too slowly and caught only 1–4 $M_\oplus$ of H/He
- **Result:** two H/He-dominated planets with metallic hydrogen dynamos, two ice-dominated planets with superionic-ice dynamos
- **Common threads:** rapid rotation, internal heat (except Uranus), rings and regular moon systems, one family of physics



Giant-planet mass evolution by core accretion at 5.2 AU: core mass (solid), envelope mass (dash-dotted) and total mass (dashed). Reproduced from Helled et al. (2014).

Rapid core assembly, a slow quasi-hydrostatic envelope contraction, then runaway gas accretion once the envelope mass exceeds the core mass.

Europa Clipper at Europa, artist's concept. Credit: NASA/JPL-Caltech.

Europa Clipper (launched 2024, ~ 50 Europa flybys) and JUICE (launched 2023, Ganymede orbit 2034) are en route to test the habitability of the Jovian ocean worlds.

Summary: Today's Course-Level Takeaways

1. **Two classes, dilute cores:** gas giants (H/He-dominated, Saturn ~ 80% H/He by mass) against ice giants (H/He only 10–20%); Juno and ring seismology reveal diffuse cores in Jupiter and Saturn
2. **Roche limit:** $d_R \approx 2.46\, R_p(\rho_p/\rho_s)^{1/3}$ bounds rings and moon accretion; tidal heating maintains the oceans of Europa and Enceladus
3. **Under-explored ice giants:** tilted dynamos powered by superionic ice, and the closest analogues of the sub-Neptunes of Lecture 13

Next: Lecture 12

Questions?

Next: **Lecture 12. Small Bodies: Meteorites, Asteroids, Comets**

Lecture notes: `formingworlds.github.io/IntroductionToPlanetaryScience`